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| **PI Name:** | Lueptow, Richard M Sc.D. | | |
| **Project Title:** | Rotating Membrane Water Purification for Long-Term Human Presence in Space | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | ADVANCED HUMAN SUPPORT TECHNOLOGIES | | |
| **Program/Discipline--Element/Subdiscipline:** | ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support | | |
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| **Space Biology Cross-Element Discipline:** | None | | |
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The recovery of potable water from wastewater is critical for life support and the environmental health of crew members during long term space missions. The wastewater stream is contaminated with particulates, microorganisms, detergents, salts, and organic compounds, all of which must be removed before the water is suitable for reuse. Standard terrestrial techniques for water treatment and recovery are not suitable for aboard spacecraft. However, rotating reverse osmosis (RO) systems, the focus of this research, show promise for water treatment and recovery. Rotating membrane systems in which a cylindrical filter rotates in a cylindrical shell setting up internal fluid flows have potential to greatly reduce the rate at which the filter loads with membrane foulants. This can substantially extend the life of water filtration systems requiring far less frequent change out of expendable components than standard filtration. Furthermore, because of their anti-fouling characteristics rotating membrane systems are applicable to ultrafiltration (UF) and RO to generate pure water. In our previous NASA-funded research, we investigated the potential for combining rotating membrane

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| **Task Description:** | filtration with RO to remove dissolved chemical species. Our initial measurements of rejection and flux for rotating RO membranes and our theoretical model indicate that rotating RO has higher flux and rejection than standard RO systems while simultaneously achieving a greater recovery. Our previous work established the concept of a low-pressure rotating RO membrane system. A mathematical model based on the film theory and the solution-diffusion model agreed well with the experimental results obtained using this first generation prototype (Lee and Lueptow 2002). Based on the model, the filtrate flux and contaminant rejection depend more strongly on the transmembrane pressure than on the rotational speed (Lee and Lueptow 2001). The first generation apparatus was built to provide proof of the concept for rotating RO and was not capable of operating at high pressures or for long time periods. Therefore, the goal of our current work has been to improve the flux of the device by increasing the transmembrane pressure up to 500 psi and enabling long term testing (7 to 60 days) of the device. To accomplish this, a second generation prototype of a rotating reverse osmosis filter and the accompanying flow circuit was designed and fabricated. The solution diffusion thin film modeling of the system was also continued. Due to the need for long-term testing with high pressures, the operational mode of the second generation apparatus was altered. The theoretical model was used to determine the optimal design geometry of the second generation prototype.<br>To further enhance the second generation prototype, the performance of commercially available reverse osmosis membranes was evaluated using a laboratory scale batch filtration method. The permeate flux and rejection of ionic solutes and dissolved organic carbon were measured experimentally to select best performing membranes among seven types of membranes (five RO and two nanofiltration) with three different ersatz wastewater solutions simulating transit mission wastewater, early planetary base wastewater, and biological water process effluent. From these screening tests, two RO membranes having both high rejection and flux characteristics are being investigated for a rotating reverse osmosis system in space applications<br><br>A separate supplemental study was conducted using ion exchange textiles as organic supports for urease immobilization with the aim of developing reactive fibrous materials able to promote urea removal. We have shown that urease can be successfully immobilized on cation exchange textiles using various coupling agents. The urease immobilized onto cation exchange textile offers various advantages. First, the textile (nonwoven fabric) form offers high surface-to-volume ratio and packing density. Furthermore, ion exchange resins in the form of beads move around resulting in abrasion which produces fine particles and degrades the surface. Using an ion exchange textile can prevent such abrasion maintaining the active surface area and preventing plugging downstream. Second, the immobilization procedure used here is simple to carry out without the need for complicated immobilization methods or the use of electron beam or gamma irradiation. Third, the activation energy of the urease immobilized is nearly as low as that of free urease, suggesting that the enzymatic reaction of the immobilized urease is nearly as good as that of the free urease. Fourth, although the kinetic parameters are slightly less favorable than those of free urease, the good storage stability and operational stability of the immobilized enzyme suggest superior potential of the immobilized enzyme for practical application. Finally, a preliminary study of the performance of the biotextile in a continuous flow reactor indicates that the biotextile can convert over 80 % of the urea over a period of several days.<br><br>The objective of this project is to advance rotating membrane technology for practical application in wastewater recovery and water purification for long-term human space flight. The effective recovery of wastewater during long-term space missions is critical for life support and the environmental health of crew members. Standard terrestrial techniques for wastewater treatment are not suitable because of microgravity or limitations related to mass, volume, and power. Except for the problem of fouling, membrane technologies are quite promising because of their compactness, reliability, and energy-efficiency. The use of rotating membrane technology seems likely to overcome the problem of fouling. Rotating filters offer the potential for smaller size than comparable cartridge filters because of their anti-fouling character. As a result, the mass and volume penalties for rotating membrane systems are most likely less than for standard membrane systems. The results of this research will make available the design tools and technology that along with the understanding of the pertinent physical processes gained in our previous research should allow rotating membrane technology to provide a significant, long-term impact in wastewater recovery and many spin-off applications. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This novel filtration system has potential use on Earth for water purification including the separation of microorganisms such as cryptosporidium from drinking water. Additional potential filtration applications include removal of fibers for the recycling of dry cleaning fluids, filtration of spent cutting oil, removal of suspended solids in wastewater such as that from laundries or industrial processes, filtration of paint chips from high-pressure water spray paint removal run-off, removal of suspended toxic heavy metals from plating and battery wastewater, and recovery of catalysts or precious metals from industrial liquid streams. |
| **Task Progress:** | Task 1 – Development of a Second Generation Prototype: Completed. A new rotating reverse osmosis filtration system was developed to accommodate an operating pressure of 500 psi for long term tests.<br>Task 2 – Advanced Design Issues for the Relevant Environment on Board a Spacecraft: Ongoing. We are working on minimizing mass, volume, and energy consumption of the rotating membrane unit, determining the scale-up factors, and extending the technology for the microgravity environment.<br><br>Task 3 – Integration of Rotating RO and Rotating MF with other Process Systems: Ongoing. We have developed a rotating microfiltration system for pretreatment prior to the rotating RO system. In addition, the Equivalent System Mass (ESM) of the life support system using rotating membrane system has been preliminarily considered. |
| **Bibliography Type:** | Description: (Last Updated: 06/28/2021) |
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| **Presentation** | Yoon, Y.; Pederson, C.; Lee, S.; Neal, R.; Lueptow, R. M. "Rotating Reverse Osmosis for Long-Term Space Missions" IWA Specialized Conference on Water Environmental-Membrane Technology<br>Jun-2004 |
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| **Presentation** | Yoon, Y.; Lueptow, R. M. "Reverse Osmosis Filtration for Ersatz Space Mission Wastewaters" International Conference on Environmental System, Rome<br>Jul-2005 |